

1. Write the difference between implicit and explicit curve representation.
2. What are the disadvantages of the Lagrange polynomial in drawing curves?
3. Why 3rd degree curve is used in Piecewise Linear Polynomial?
4. What is the degree of Parametric and Geometric continuity for this equation:
 - a. if $0 < x \leq 0.5$ then $(2x, 0)$
Else if $x > 0.5$ then $(1 + 2(x - 0.5), 2(x - 0.5))$
 - b. If $0 < x \leq 1$ then $(x, 0)$
Else if $x > 1$ then $(3(x - 1) + 1, 3(x - 1) + 1)$
5. What is the degree of Parametric continuity for this function?(check up to $C(2)$)

$$P(x) = \begin{cases} (x, -2x^3 + 3x^2) & \text{for } 0 \leq x \leq 1 \\ (x, -x^3 + 3x - 1) & \text{for } 1 < x \leq 2 \end{cases}$$

6. Derive the Matrix equation of a Quadratic Bezier Curve where 3 control Points p_1 , p_2 , and p_3 are given.
7. Suppose the control Points are:
 $P_1 = (1, 1)$
 $p_2 = ?$
 $p_3 = (4, 5)$
 $p_4 = (6, 0)$
 The Bezier curve passes through point $A(t) = (3.2, 2.68)$, where $t = 0.4$. Find the control Point P_2 .

Deadline: 18 May 11:59 PM

Form Link: <https://forms.gle/Va5nWJMdz64bdpJt9>